ECON100B - SECTION 6

The Solow Growth Model

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ANNOUNCEMENTS/AGENDA

Announcements

- PSET 1 due Friday, 2/13

Agenda

- Solow Growth Model
- Solow PSET problems (Time permitting)

Question of the Day

1. Under the Cobb-Douglas model, is Capital (K) or Total Factor Productivity (A) more important in determining the economic output of a country? What is the purpose of Total Factor Productivity in our model, and what are some factor that could potentially influence Total Factor Productivity?

2. What is your favorite sport/sporting event (or just something else on TV) to watch?

The Solow Growth Model

BIG PICTURE QUESTIONS

- **What sustains growth in the long run?**
  - Will it continue in the future?
- **Why are some countries so far behind the frontier?**
  - What might help them close the gap?

FOOD FOR THOUGHT

If you were put in charge of every aspect of a country right now and told you your only goal was to grow the economy's per-capita income, what would you focus on?

IS CAPITAL ACCUMULATION KEY TO GROWTH?

- Seems plausible - just keep investing in more productive capacity
- Conventional wisdom in the 1950s - Yes!
- **Solow (1956) - No, you can’t grow indefinitely with capital accumulation alone**

Solow Model Setup

- Consider a large farm that grows a single crop, corn
- ¾ of the corn is reserved for eating, while ¼ of the corn is saved for planting for next year
- Each saved corn kernel produces ten ears of corn, so the size of harvest can grow larger in every year
- The Solow model helps us describe and understand capital accumulation (“corn”) over time

The Production Function, Revisited

$$Y_t = F\left(K_t, L_t\right) = \bar{A}K_t^{1/3}L_t^{2/3}$$

- We add a new time subscript to the production function
- In the model economy, output can be used for one of two purposes, consumption or investment:

$$C_t + I_t = Y_t,$$

- This is a resource constraint, as the economy is constrained to spending on Consumption or Investment, equal to output

Accumulation of Capital

- We saw previously, that we saved a certain amount in investment

$$C_t + I_t = Y_t,$$

- Our future capital is determined by the following equation:

$$K_{t+1} = K_t + I_t - \bar{d}K_t.$$

- What this equation says: capital next year is equal to capital this year added to investment this year, minus the depreciation of capital
- We use an exogenous depreciation rate, commonly 7% or 10%

Accumulation of Capital

Time, t	**Capital, K_t**	**Investment, I_t**	**Depreciation, $\bar{d}K_t$**	**Change in capital, ΔK_{t+1}**
0	1,000	200	100	100
1	1,100	200	110	90
2	1,190	200	119	81
3	1,271	200	127	73
4	1,344	200	134	66
5	1,410	200	141	59

How do we determine investment?

- **Investment:** $S = I = sY$
- **Consumption:** $C = (1 - s)Y$

- We set "s" as the fraction of our output that is set aside for investment
- "1-s" represents the fraction of our output that is set aside for consumption
- By logic, s + (1 – s) has to be equal to 1, where s can be any number between 0 and 1

The real interest rate

- The real interest rate is the amount that a person can earn by saving one unit of output for a year, or equivalently the amount a person must pay to borrow one unit of output for a year
- The real interest rate equals the rental price of capital that clears the capital market, which in turn is equal to the marginal product of capital
- Why? Savings (“S”) = Investment (“I”) which becomes the capital supplied in the economy

Output per person

Recall from the Cobb Douglas model:

Output per person as:

$$y_t = Y_t / L_t$$

Capital per person:

$$k_t = K_t / L_t$$

$$\begin{aligned} y_t \equiv \frac{Y_t}{L_t} &= \frac{\bar{A}K_t^{1/3}L_t^{2/3}}{L_t} \\ &= \frac{\bar{A}K_t^{1/3}}{L_t^{1/3}} \\ &= \bar{A}k_t^{1/3}. \end{aligned}$$

Output per person

- Solving for output per person tells us that output per person is dependent on capital
- To understand how output per person (y_t) changes over time, we need to understand how capital per person (k_t) changes over time

$$\underbrace{\Delta K_{t+1}}_{\text{change in capital}} = \underbrace{\bar{s}Y_t - \bar{d}K_t}_{\text{net investment}}.$$

- The change in capital stock is equal to investment minus depreciation, a.k.a net investment

$$\underbrace{\Delta k_{t+1}}_{\text{change in capital per person}} = \underbrace{\bar{s}y_t - \bar{d}k_t}_{\text{net investment per person}}.$$

The Solow Model

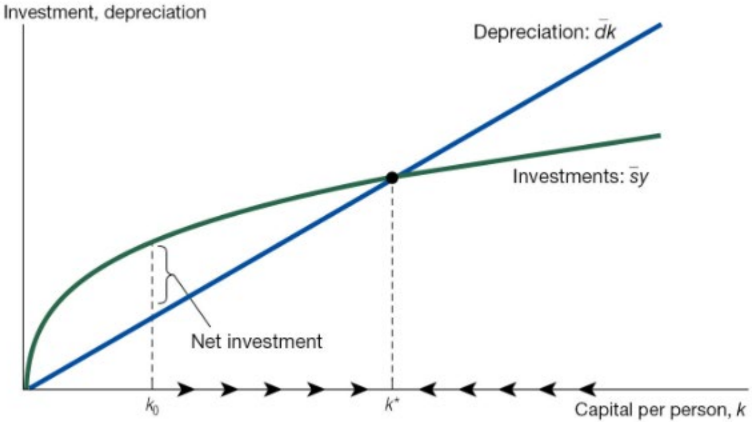
- The essence of the Solow model is understanding of output per person changes over time (dependent on capital) and how capital changes over time:

$$y_t = \bar{A} k_t^{1/3}, \tag{5.6}$$

$$\Delta k_{t+1} = \bar{s} y_t - \bar{d} k_t. \tag{5.7}$$

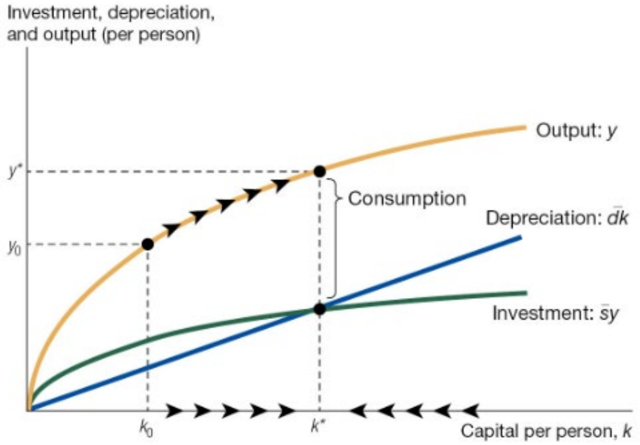
- We cannot solve these equations algebraically (too many endogenous variables) but can analyze it in a graph

The Solow Diagram



- Where our investment curve can be written as $\bar{s}y = \bar{s}\bar{A}k^{1/3}$.

The Solow diagram with output



Solow Model Practice

1. If a country's total factor productivity (A) stays the same in the long run, and savings rates and depreciation rates remain constant, what will happen to output per capita in the economy in the long run? What will happen if we increase the savings rate?
2. If an economy's capital per capita is lower than the "steady state", will economic growth be faster or slower as the economy approaches the "steady state"?
3. If my output per capita (y0) is 50 units, and my savings rate is 20%, while my current capital stock per capita (k0) is 100 units and my depreciation rate is 10%, what will happen to my output per capita (y1) next year, if A (TFP) stays constant?

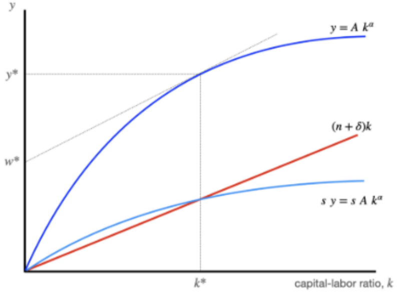
PSET Questions

3. [16 points total] So how low can we go, Solow? A capital response to immigration?

In the standard neoclassical growth model of Robert Solow without technological growth, consumers supply labor inelastically and save out of income at rate s. There are constant returns to scale in labor and capital, and productivity A is fixed and not changing. Capital depreciates at rate δ, and labor increases at the rate of population growth, n. When output is Cobb-Douglas,

capital per worker is	$k = K/L$	also called the "capital-labor ratio"
output per worker is	$y = A\,k^{\alpha}$	and typically $\alpha = 1/3$
saving per worker is	$s\,y = s\,A\,k^{\alpha}$	
depreciation per worker is	$(n + \delta)k$	

The Solow diagram below shows how capital per worker reaches a *steady state* at k*, where depreciation per worker equals saving per worker: $(n + \delta)k = s\,y$, so capital per worker is unchanging there. The steady state is shown by the vertical dashed line in the diagram below.



PSET Questions

(a) [2 points] If the capital per worker is in steady state at k^* and not changing over time, what is true about the wage w^*, which is shown in the graph on the y-axis?

PSET Questions

Suppose that the economy is in steady state, and then the government carries out a large removal of immigrant workers, one that increases capital per worker k by about 25%. The policy removes workers but leaves capital K and productivity, A, unchanged.

(b) [4 points] Illustrate this policy by augmenting the graph from part (a) as needed. Draw any new curves and show jumps and changes in variables. What happens to the wage in the short run? What happens to the wage in the long run? Explain fully. [Hint: do any of the parameters like n, δ, s, A, and α change?]

(c) [2 points] A higher level of capital per worker will produce a higher wage, because workers are more productive. And removing any type of worker will raise capital per worker. But is that change temporary or permanent? Why?

PSET Questions

Now suppose that while in steady state, the government instead decides to change **immigration policy**, reducing number of refugee and work visas each year, which reduces the rate of population growth, n, to a new level $n' < n$. (Assume there is no removal policy.)

(d) [4 points] Illustrate this policy by augmenting the graph from part (a) as needed. Draw any new curves and show jumps and changes in variables. What happens to the wage in the short run? What happens to the wage in the long run? Explain fully. [Hint: do any of the parameters like n, δ, s, A, and α change?]

(e) [4 points] What would have happened in the graph in pa rt (d) if immigration enhanced productivity, A, so that reducing immigration reduced the level of productivity? Explain.

The Steady state

- While we can't solve for capital at any given point, we are able to solve for the steady state level of capital:

$$\bar{s}y^* = \bar{d}k^*.$$

$$\bar{s}\bar{A}k^{*1/3} = \bar{d}k^*.$$

$$k^* = \left(\frac{\bar{s}\bar{A}}{\bar{d}}\right)^{3/2}.$$

Steady State Output

- Associated with a steady-state level of capital is a steady state level of production

$$y^* = \bar{A}k^{*1/3}.$$

- Thus, we achieve the following expression for steady-state production if we plug in k* into the equation:

$$y^* = \left(\frac{\bar{s}}{\bar{d}}\right)^{1/2} \bar{A}^{3/2}.$$

Growth across countries

- Empirically, different countries have different investment rates:
- For example, investment in South Korea has averaged 35% of GDP, investment in United States has been around 25%, while in poorer countries, it's around 10%
- Empirically, we observe that investment rates in the richest countries average 25-30% while investment rates in poor countries are around 7%
    - Based on our model, differences in investment rates explain a factor of 2 times the differences in income, while 37.5 is explained by the productivity differences (!)

Lessons from the Solow Model

- In the long run, diminishing returns to capital accumulation cause investments to fall
- Eventually, depreciation and new investment offset each other and we reach steady state
- Capital accumulation cannot serve as the engine of long-run economic growth